# Assignment 3 · Ground Truth

Alewife Brook Reservation, Cambridge MA
the bank of the Little River beside the constructed stormwater wetland (2013)

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Field visits Oct 2026 · pin-up Fri Oct 30 2026, 12 pm

## SHEETS


**PIN-01** Part A · Patch Delineation (group delineation)
**PIN-02** Part B · Ground-Truthed Patch Map
**PIN-03** Part C · Plot + Transect
**PIN-04** Part C · Plants
**PIN-05** Part C · Field Notes + Observations
**PIN-06** Site Photos
**PIN-07** Nest · Habitat Feature
**PIN-08** Scan + Splat Renders

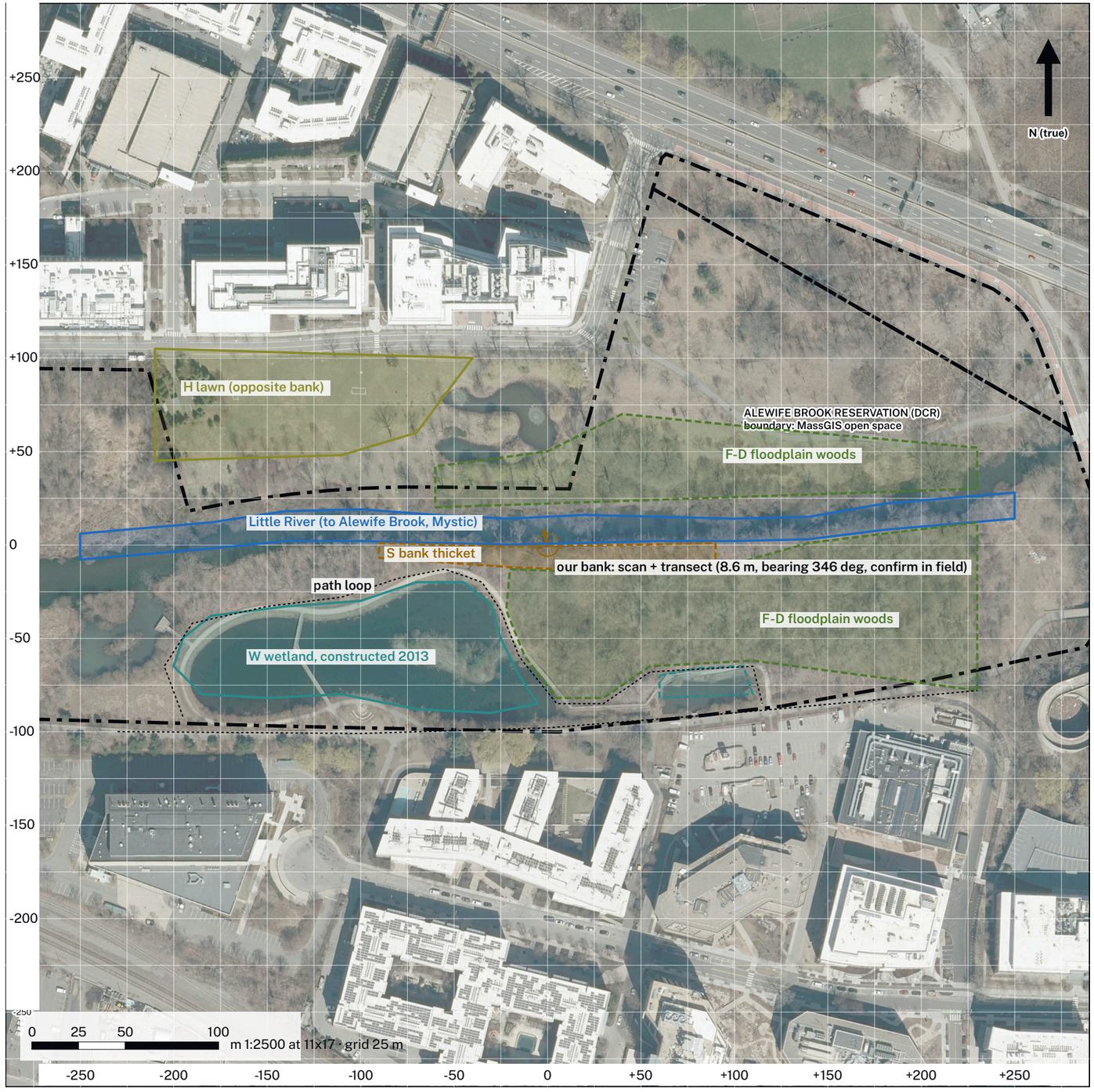

Estimated values are marked (grey italic); grey italic notes = confirm in field.

*Abandoned bird's nest with woven plastic wrappers, found on the bank (PIN-07).*

# Part A · Patch Delineation (group delineation)

# PIN-01

Group delineation on the MassGIS 2023 leaf-off aerial at 1:2500, from the group's plant catalog and field observations (Oct 2026). Solid = edge seen on the aerial and on site; dashed = uncertain.




## KEY group delineation (spec A1 classes)

| | | |
|---|---|---|
| O | O | open water: Little River |
| W | W | wetland: constructed stormwater wetland (2013) |
| S | S | shrub: bank thicket (brambles, buckthorn, knotweed) |
| F-D | F-D | woody, deciduous: floodplain woods |
| H | H | upland herbaceous: manicured lawn (opposite bank) |
| path | path | path: Alewife Brook Pathway loop |

Solid edge = seen on the aerial and on site; dashed edge = uncertain. Not found in our area: F-E evergreen, B bare ground (except the compacted bank), P paving (outside the reservation).

## METHOD

Edges traced by eye on the leaf-off aerial where tone or texture changes over about 10 m (minimum patch about 100 m2), then checked against what we saw on the bank (Oct 2026) and the species in the group's plant catalog. The catalog lists species, not positions: knotweed and buckthorn are placed in the bank thicket and disturbed openings, goldenrod at edges, apple and sugar maple in the floodplain woods (approximate). Patches outside our transect are from the aerial only.

## FIELD NOTES (observations, Oct 2026)

· Our bank: the south / southwest-facing bank of the Little River, between the river (to Alewife Brook, then the Mystic) and the Alewife Brook Pathway loop.

· Fallen telephone pole across two oaks: onto one mature oak, catching a second at its Y-branch. Moss on pole and oaks, no wires: likely years ago. The gap let saplings in.

· Soil: compacted bank; sticky mud in the wetland bog; dark silty detritus about 6 in over gray clay at about 1 ft; no rotting smell.

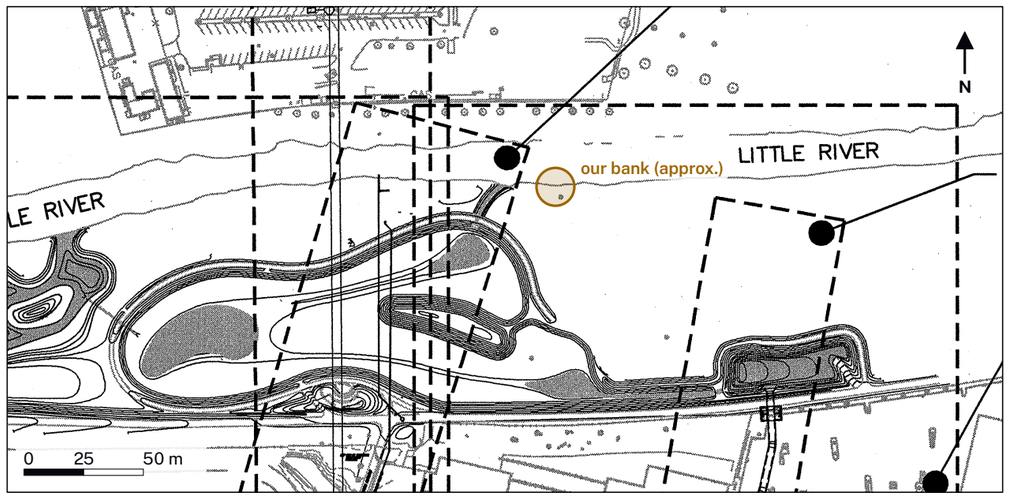
· Water: shallow and still, about 0 to 1 ft at the margin, 2 to 3 ft further out; dead trees and branches break the surface.

· Brambles and bush thickets on the bank and in the wetland. Human use: an encampment, waste, plastic.

· Opposite bank (N / NW): manicured lawn, so possible fertiliser and stormwater runoff into the river.

· Two abandoned bird's nests with plastic wrappers woven in (PIN-07). Forebay cleaned out with a front-end loader; city and state land (DCR / DEP).

## SITE BLUEPRINT excerpt




**Excerpt, City of Cambridge construction documents, Alewife Reservation stormwater wetland.**

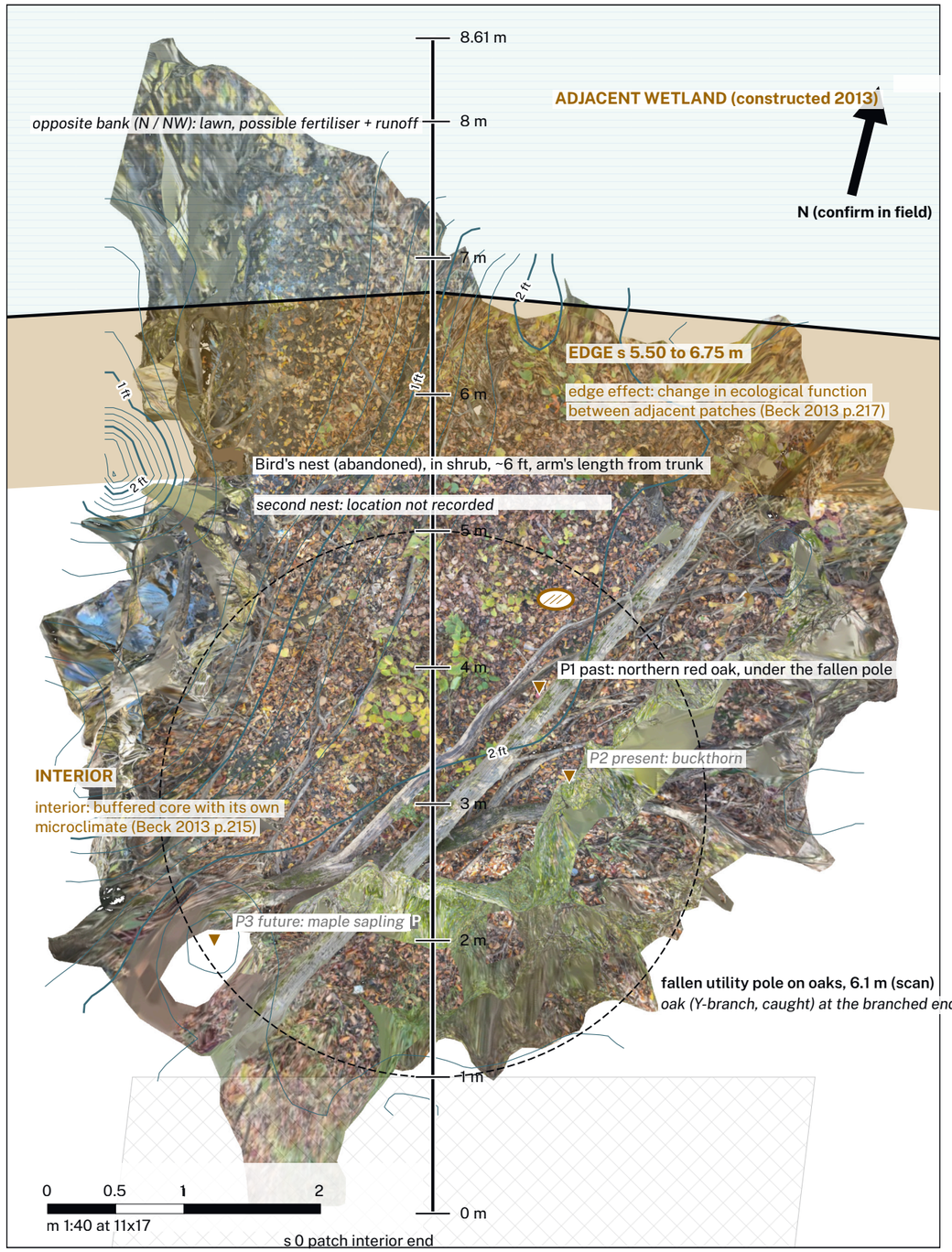
Sheet G-1, Drawing Index and Key Plan, Conformed Set, Sept 1 2010. Drawn 1 in = 200 ft, shown at 1:2500 like the aerial.

Aerial: MassGIS orthos2023, spring 2023 leaf-off, 15 cm (tiles.arcgis.com/.../orthos2023/MapServer). Boundary: MassGIS Protected and Recreational OpenSpace (gis.eea.mass.gov/server/rest/services/Protected_and_Recreational_OpenSpace_Polygons), Alewife Brook Reservation, DCR. Scan location 42.396871, -71.146994.

# Part B · Ground-Truthed Patch Map

PIN-02

Part B. Group patch on the scan footprint: Polycam mesh texture, top view, ortho, plan 1:40 at 11x17. Fallen utility pole on oaks across the plot; species from the group's plant catalog.




## KEY plot layers

- base: Polycam mesh texture, top view, ortho
- ground contours 0.25 ft (scan ground grid)
- index contour 1 ft, labelled (scan z datum)
- edge: functional band 1.25 m (s 5.50 to 6.75)
- group patch boundary (estimated outline)
- adjacent patches, spec A1 classes (estimated)
- transect, ticks 1 m, s 0 at the patch interior
- fixed-radius plot r 2.0 m
- P1 past (oak) · P2 present · P3 future
- bird's nest (abandoned), position to confirm

12.3 m measured (scan / field)

**Gaussian splat, perspective A (Polycam scan, trained locally)**

*Grey italic = estimated, not measured. Ochre = terms (reading, page).*

## SITE NOTE

Constructed stormwater wetland, 3.4 ac, opened Oct 15 2013; it treats stormwater before it reaches the Little River and cuts combined sewer overflows to Alewife Brook (City of Cambridge handout p.1 to 2; overview p.1 to 2).

## OBSERVED (Oct 2026)

Disturbance: fallen utility pole on two oaks (canopy gap, saplings in, legacy wood); brambles and thickets on the bank; soil compaction; human use (an encampment, waste, plastic). Adjacency: manicured lawn on the opposite bank (N / NW), possible fertiliser and runoff.

## TERMS (readings + ett3 lectures)

patch: an area that differs from what surrounds it (Beck 2013 p.211; ett3 lecture, Oct 9 2026)
edge effect: change in ecological function between adjacent patches (Beck 2013 p.217)
disturbance: a process, not just an event: the pole fall opened the canopy (ett3 lecture, Oct 2 2026)
ruderals: fast growers that take disturbed openings (knotweed, buckthorn) (ett3 lecture, Sep 11 2026)

## PLANT TABLE primary species from the group's plant catalog · companions: oak, pole, nests

| common | Latin | form / str. | patch | wetl. ind. | origin / MA | role |
|---|---|---|---|---|---|---|
| **Japanese knotweed** | *Reynoutria japonica* | herb, tall / U | bank thicket, disturbed edge | FACU | E Asia; invasive, prohibited in MA | edge |
| **European buckthorn** | *Rhamnus cathartica* | shrub / M | canopy gap, bank thicket | FAC | Europe; invasive, prohibited in MA | P2 present |
| **Goldenrod** | *Solidago spp.* | herb / G | edge, openings | FACU to FACW | native | |
| **Sugar maple** | *Acer saccharum* | tree / O (sapling U) | floodplain woods, gap | FACU | native | P3 future |
| **Domestic apple** | *Malus domestica* | tree / M | floodplain woods edge | UPL, confirm | Central Asia; non-native, not listed | |
| **Northern red oak** | *Quercus rubra* | tree / O | under the fallen pole | FACU | native | P1 past |
| **Oak, caught at Y-branch** | *Quercus sp.* | tree / O | under the pole | | native | companion |
| **Fallen utility pole** | *not a plant* | legacy wood / G | across the plot | | | companion |
| **Bird's nests (2)** | *habitat feature* | shrub, ~6 ft | s 4 to 5 m; 2nd not recorded | | | habitat |

Indicator: USACE National Wetland Plant List, Northcentral and Northeast. MA invasive: MDAR Prohibited Plant List. Silver maple (A. saccharinum, FACW) is the floodplain maple expected here; confirm against the sugar maple ID. Earlier tentative photo IDs are on REF-03 only.

## MARSH ZONATION planted palette per City of Cambridge (designed planting; not observed by us)

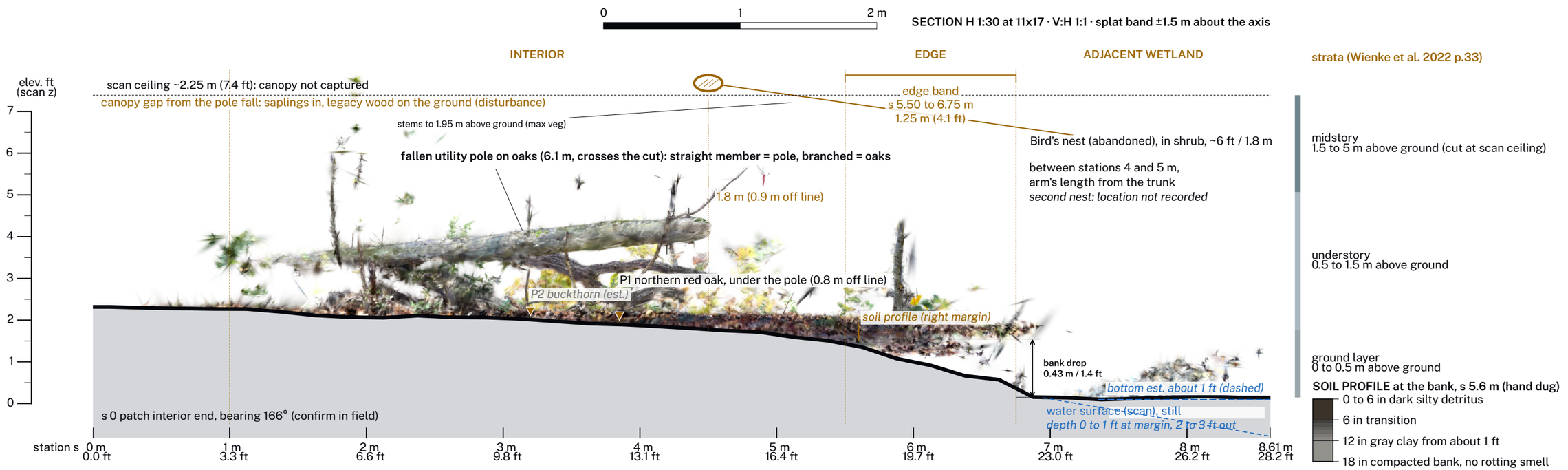
| | | |
|---|---|---|
| **deep marsh** | 1 to 1.5 ft | hardstem and softstem bulrush, lesser bur-reed |
| **open water** | 1.5 to 3 ft | white water lily, pickerelweed, northern arrowhead |
| **emergent marsh** | 6 in to 1 ft | arrow arum, spike rush, green bulrush, blue flag iris |
| **high marsh** | up to 6 in | tussock sedge, marsh marigold, riverbank wild rye, Canada rush, wool grass, marsh hibiscus |
| **scrub-shrub** | bank | silky dogwood, pussy willow, black chokeberry, sweet pepperbush, meadowsweet, highbush blueberry |

*Reference gradient for the wetland patch: deep to high marsh, then scrub-shrub at the bank.*

# Part C · Plot + Transect

**PIN-03**

Part C composite. Transect section on the gaussian splat side view, band ±1.5 m, H 1:30, V:H 1:1, stations every 1 m, s 0 at the patch interior. Plant catalog, sampling table per zone, plot in plan.



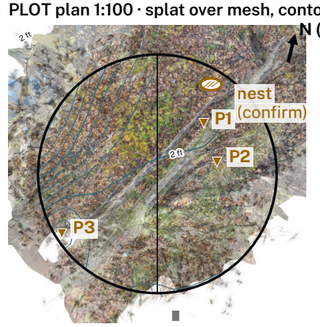

Edge vs interior (functional): the edge band, s 5.5 to 6.75 m, is where light, moisture and species change between patches; the interior is the buffered core with its own microclimate (Beck 2013 p.215 to 217; PIN-02).

## PLANT CATALOG *species on our transect, from the group's plant catalog*

| species | form / stratum | where on the transect | wetland ind. | origin / MA status | role |
|---|---|---|---|---|---|
| **Japanese knotweed** *Reynoutria japonica* | herb, tall / U | bank thicket, disturbed edge | FACU | E Asia; invasive, prohibited in MA | edge, disturbance |
| | Dense stands to about 10 ft that shade out natives; spreads by long rhizomes; rivers, roadsides, rail corridors, disturbed urban sites. | | | | |
| **European buckthorn** *Rhamnus cathartica* | shrub / M | canopy gap, bank thicket | FAC | Europe; invasive, prohibited in MA | P2 present |
| | Introduced as a hedge plant; birds eat and spread the black berries; dense shade suppresses native understory. | | | | |
| **Goldenrod** *Solidago spp.* | herb / G | edge, openings | FACU to FACW | native | |
| | Many species in MA; blooms late summer to fall; key nectar source for bees and butterflies before winter. | | | | |
| **Sugar maple** *Acer saccharum* | tree / O (sapling U) | floodplain woods, gap saplings | FACU | native | P3 future |
| | Signature New England tree; source of maple syrup; can live 300+ years; food and habitat for birds and squirrels. | | | | |
| **Domestic apple** *Malus domestica* | tree / M | floodplain woods edge | UPL, confirm | Central Asia; non-native, not listed | |
| | Spring blossoms for pollinators; fruit for birds, mammals, people; feral apples common in old New England fields. | | | | |
| **Northern red oak** *Quercus rubra* | tree / O | under the fallen pole | FACU | native | P1 past |
| | ID from form and site, confirm. | | | | |

Indicator: USACE National Wetland Plant List, Northcentral and Northeast. MA status: MDAR Prohibited Plant List. Silver maple (A. saccharinum, FACW) is the floodplain maple expected here; confirm against the sugar maple ID.

**PLOT plan 1:100 · splat over mesh, contours 0.25 ft**




*r 2.0 m, centre s 3.0 m (confirm)*

**KEY**

- ground section line (scan ground, 5th pct z)
- below ground (poche)
- water surface (scan), depth observed
- scan ceiling 2.25 m
- edge band, zone breaks

**Grey italic = estimated, not measured.**

## SAMPLING TABLE one column per zone (spec C2 to C4)

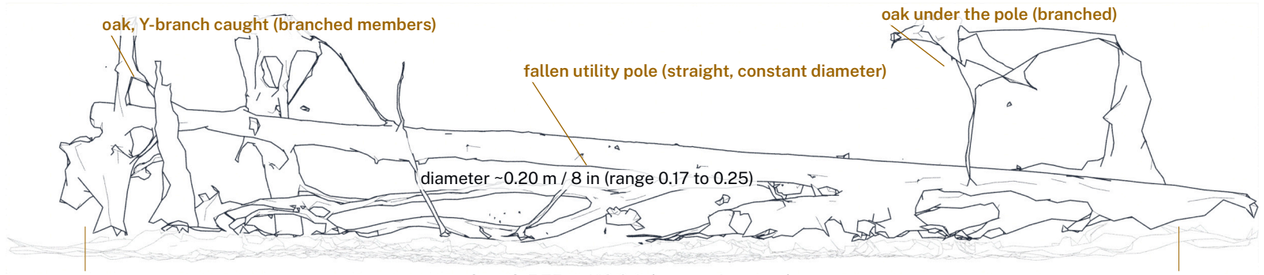
| | **interior** s 1 to 5.5 m | **edge** s 5.5 to 6.75 m | **wetland** s 6.75 to 8.61 m |
|---|---|---|---|
| **dominant species** | *R. cathartica* | *A. saccharum* | *Carex sp.* |
| **% cover** | *71* | *68* | *78* |
| **no. of species** | *6* | *7* | *7* |
| **veg. height max (scan)** | 1.95 m (6.4 ft) | 1.07 m (3.5 ft) | 0.91 m (3.0 ft) |
| **strata present (scan)** | U G | U G | U G |
| **canopy (densiometer)** | *shade, 22 % open* | *partial, 32 % open* | *open, 63 % open* |
| **woody presence** | *saplings, seedlings, vines* | *saplings, vines* | *seedlings* |

Grey italic = estimated, not measured. Height and strata from the scan (strata by height above ground; no returns above 2.25 m). Canopy: shade < 25 % open, partial 25 to 60 %, open > 60 %.

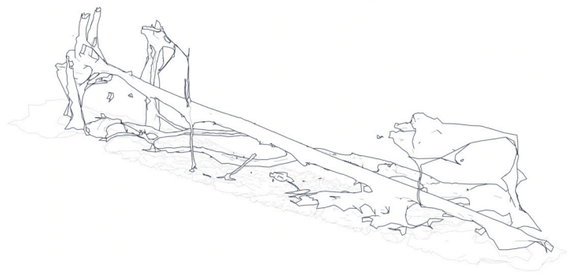
# Part C · Plants

**PIN-04**

Part C5. Three plants: P1 past, P2 present, P3 future. P1 = northern red oak under the fallen utility pole, line drawing from the scan mesh (Blender Freestyle, ortho side view at 1:30). P2 / P3 from the group's plant catalog.

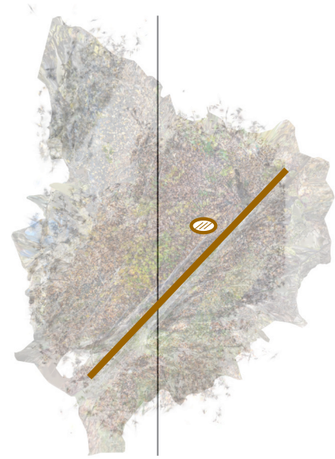



SIDE ELEVATION 1:30 at 11x17 · looking across the log axis · line weights: outline / creases / ground



3/4 VIEW not to scale

INSET pole on the plan, transect, nest



## P1 · PAST Northern red oak (Quercus rubra), under the fallen utility pole. ID from form and site, confirm.

Measured (scan): the long straight member of constant diameter is the pole: length 5.75 m (18.9 ft); diameter about 0.20 m (8 in), range 0.17 to 0.25 m; top 0.66 m above ground at the edge end, 0.22 m at the interior end; ground contact about 1.0 m, the rest rests on the caught oak. The curving branched members are the oaks. Not measured in the field: butt diameter, root plate, decay class.

What happened: a telephone pole (no wires, moss on it, so likely years ago) fell onto one mature oak and caught a second oak at its Y-branch, taking it down too.

Now: legacy wood on the ground, moss on the pole and the oaks; a canopy gap overhead with saplings coming in (disturbance-driven succession).

Before: closed oak canopy on the bank. After (expected): the gap fills with saplings; invasive buckthorn and knotweed can take the opening first (hypothesis).

Companion element, the pole: a disturbance agent, not a plant. Wood and preservative treatment speculative; read the brand / tag in the field.

## P2 · PRESENT

**European buckthorn (Rhamnus cathartica), invasive: colonising the gap**

*from the group's plant catalog*

· Introduced as a hedge plant.
· Birds eat and spread the black berries.
· Dense shade suppresses native understory.

Hypothesis: invasives like buckthorn and knotweed thrive in disturbed openings like the pole-fall gap (ruderals, ett3 lecture, Sep 11 2026).

Indicator FAC (NWPL NCNE). Prohibited from sale and distribution in MA (MDAR).

*photo: the group's plant catalog*

## P3 · FUTURE

**Sugar maple sapling (Acer saccharum): canopy return**

*from the group's plant catalog*

· Signature New England tree.
· Source of maple syrup.
· Can live 300+ years.
· Food and habitat for birds and squirrels.

Saplings in the gap grow toward the canopy the pole opened (succession, ett3 lecture, Sep 11 2026).

Indicator FACU (NWPL NCNE). Silver maple, A. saccharinum, is the floodplain species expected here; confirm.

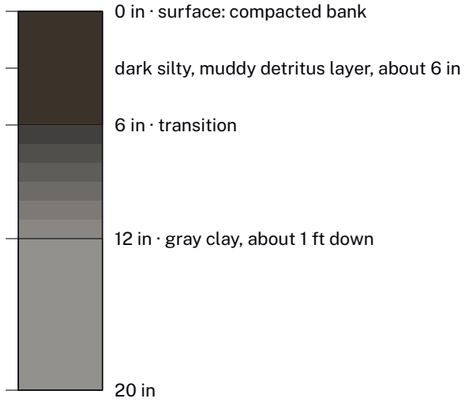
*photo: the group's plant catalog*

# Part C · Field Notes + Observations

# PIN-05

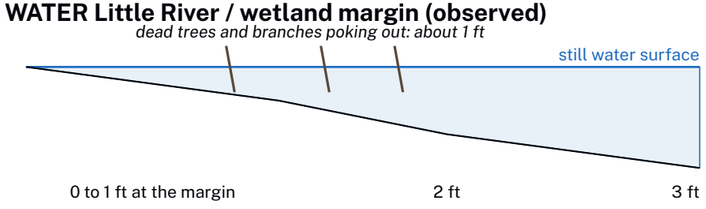
Field observations from the group's site visits, Alewife Brook Reservation, Oct 2026, as an annotated sheet, with a section drawn from the Polycam scan.

## SOIL PROFILE at the bank (observed)




Sticky mud in the wetland bog. No rotting smell detected. Fills the spec soil-horizons item; depths by eye and hand, not measured with a probe.

## WATER Little River / wetland margin (observed)




Shallow, still, slow flow from the wetland forebay toward the Little River, then Alewife Brook and the Mystic. Bottom of fine silt, sand and mud (habitat for burrowing worms, snails, aquatic insects, freshwater mussels). Section not to scale.

## DISTURBANCES (spec C1)

· Fallen telephone pole on two oaks: onto one mature oak, catching a second at its Y-branch. Moss on both, no wires: likely years ago.
· Canopy gap from that fall: saplings in (disturbance-driven succession).
· Brambles and bush thickets on the shallow bank and in the wetland.
· Soil compaction on the bank.
· Human use: an encampment, human waste, plastic.

## NESTS (habitat feature)

· Two abandoned bird's nests with plastic wrappers woven into the structure: deliberate, not blown in. Adaptive reuse of plastic litter, possibly as weatherproofing.
· Nest 1: in a shrub, about arm's length from the trunk, about 6 ft up, between stations 4 and 5 m.
· Nest 2: location not recorded. Both collected (PIN-07).

## ADJACENCY

· Our bank faces south / southwest, between the Little River and the Alewife Brook Pathway loop.
· Opposite bank (N / NW): manicured lawn, so possible fertiliser and stormwater runoff.

## MANAGEMENT

· The wetland forebay is cleaned out with a front-end loader (the City's term: "objectionable solids").
· The City monitors wetland habitat, but infrequently. Ownership: a mix of city and state land, overseen by DCR / DEP.
· Constructed stormwater wetland, 3.4 ac, opened Oct 15 2013 (City of Cambridge handout p.1 to 2).

## MARSH ZONATION

*planted palette per City of Cambridge (not observed by us)*

| | |
|---|---|
| **deep marsh**<br>1 to 1.5 ft | hardstem and softstem bulrush, lesser bur-reed |
| **open water**<br>1.5 to 3 ft | white water lily, pickerelweed, northern arrowhead |
| **emergent marsh**<br>6 in to 1 ft | arrow arum, spike rush, green bulrush, blue flag iris |
| **high marsh**<br>up to 6 in | tussock sedge, marsh marigold, riverbank wild rye, Canada rush, wool grass, marsh hibiscus |
| **scrub-shrub**<br>bank | silky dogwood, pussy willow, black chokeberry, sweet pepperbush, meadowsweet, highbush blueberry |

## TERMS ett3 lectures (class notes)

patch: an internally homogeneous area that differs from its surroundings (ett3 lecture, Oct 9 2026)

edge structure: abrupt, feathered, gradual edge; forest interior (ett3 lecture, Oct 9 2026)

disturbance: a force or process causing a perturbation relative to a reference state; a process, not just an event (ett3 lecture, Oct 2 2026)

succession: a continual shifting of conditions within an environment (ett3 lecture, Sep 11 2026)

ruderals (CSR): fast growing, many seeds, germinate in many conditions; non-native is not the same as invasive (ett3 lecture, Sep 11 2026)

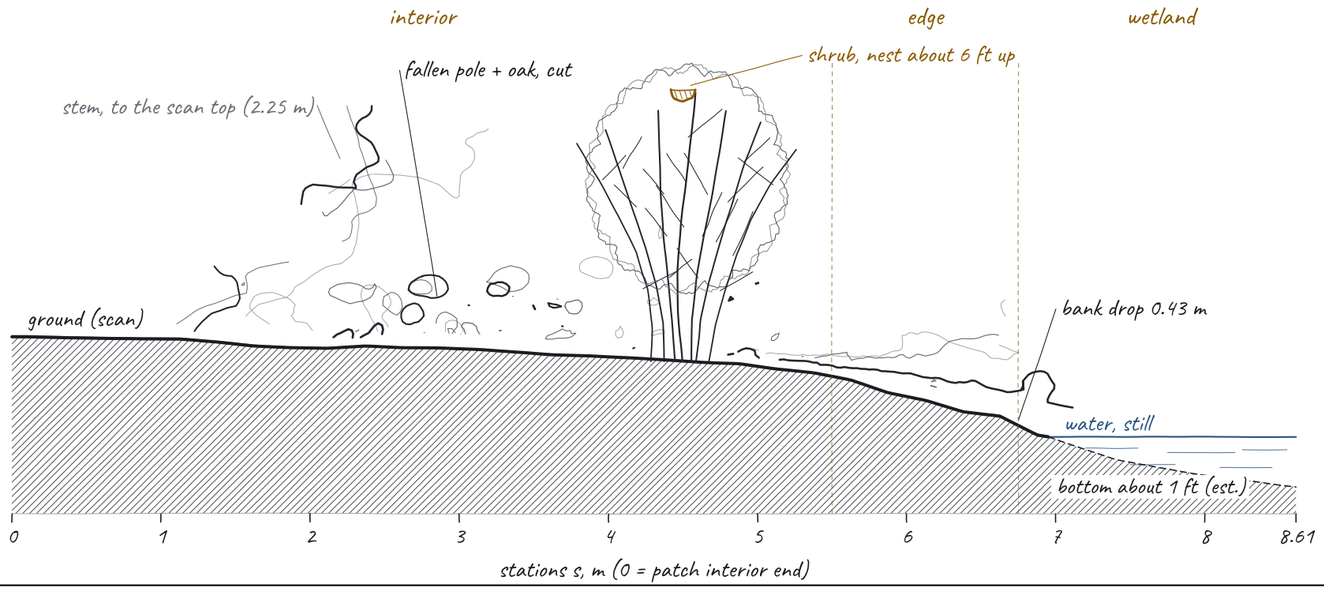
forest stand dynamics: disturbance opens pockets of light; stands initiate as a patch mosaic (ett3 lecture, Oct 2 2026)

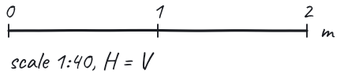
## SOURCES

Observations: group field visits, Oct 2026. City facts: City of Cambridge, Alewife Reservation Constructed Wetland project overview and opening handout (Oct 15 2013); marsh zones from the City's project page. Construction sheets in hand (title, G-1, G-3, G-4) do not give the normal pool elevation.

## SECTION DRAWN FROM THE POLYCAM SCAN

Line drawing generated from the scan mesh, cut along the transect (the PIN-03 cut), with lines beyond at 0.5 and 1.0 m. Not a field sketch. The scan does not resolve the shrub and nest: they are drawn at their recorded position.




## LINES

- the cut, t 0 (pen)
- beyond, t ±0.5 m
- beyond, t ±1.0 m
- ground (scan, 5th pct z)
- water surface (scan)
- bottom, estimated
- edge band, s 5.5 to 6.75 m

Hatch = ground body (poche). Labels in a hand font (Caveat, OFL).

# Site Photos

PIN-06

Top: photos of the primary species from the group's plant catalog. Bottom: keyframes from our scan walk of the bank, Oct 16 2026, matched to the catalog species where they show them.

## PLANT CATALOG photos of the primary species

**Japanese knotweed, Reynoutria japonica**
invasive, prohibited in MA

**European buckthorn, Rhamnus cathartica**
stems, understory · invasive

**European buckthorn, Rhamnus cathartica**
black berries, bird dispersed

**Goldenrod, Solidago spp.**
fall seed head · native

**Sugar maple, Acer saccharum**
native

**Domestic apple, Malus domestica**
fruit and leaves

## OUR BANK scan-walk keyframes, Oct 16 2026 (no separate field photos received)

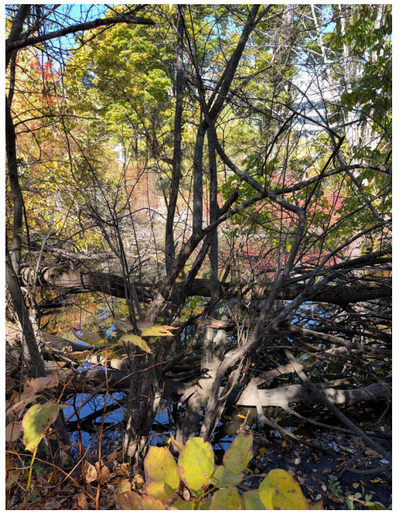
**Buckthorn, multi-stem shrub, interior**
matches the catalog's European buckthorn · scan-walk keyframe, Oct 16

**Goldenrod, ground layer, edge**
matches the catalog's goldenrod · scan-walk keyframe, Oct 16

**Fallen utility pole on oaks**
moss, branches, litter; pole 5.75 m (scan) · keyframe, Oct 16

**Edge: woody stems thin out**
herbs and vines on the slope; edge band s 5.5 to 6.75 m (scan)

**Bank drop to standing water**
0.43 m (1.4 ft) (scan); compacted bank, observed



**Wetland margin: still, shallow water**
fallen branches; about 0 to 1 ft deep (observed)

# Nest · Habitat Feature

Two abandoned bird's nests from the bank, photographed in the studio. Habitat features layer (spec C1) and plastic reuse. Range of distances: wide (B), mid (A), close (C), plan (D).

**PIN-07**

*A · mid: oblique, stone behind, the full branch (hero)*

*D · plan: top-down*

*C · close: underside, woven plastic*

*B · wide: front, nest on the branch, stone base*

**Abandoned bird's nests (2) found on the bank, Alewife Brook Reservation, Oct 2026. Plastic wrappers woven into the structure. Collected and turned into a readymade sculpture.**

Habitat features layer: nest 1 sat in a shrub about 6 ft up, about arm's length from the trunk, between stations 4 and 5 m on our transect (PIN-02, PIN-03); the second nest's location was not recorded.

Plastic reuse: the wrappers are woven in, not blown onto the nest, so the birds chose the material: adaptive reuse of plastic litter from the bank, possibly as weatherproofing. The same bank carries human use (an encampment, waste, plastic), so the litter is close at hand.

## ANTHROPOGENIC NEST MATERIAL

Many birds build human-made materials into their nests: plastic, cigarette butts, even metal. A global review of 25 studies (24 species, 10,790 nests) found this use rises with the human footprint (Jagiello et al. 2019, Environmental Pollution). Proposed functions include structure and defence. In Mexico City, house sparrows and house finches lined nests with cigarette-butt fibre; nests with more fibre carried fewer ectoparasites, though the toxins may cost the birds (Suárez-Rodríguez et al. 2013, Biology Letters). Crows and magpies in Rotterdam and Antwerp have built nests from anti-bird spikes (Hiemstra et al. 2023, Deinsea). Read as behavioural flexibility and niche construction, it is animals reshaping their surroundings with what the city supplies. Our nest, with plastic wrapper woven into its walls, fits this pattern.

Anthropogenic nest material: human-made material used by animals in nest building.

Niche construction: organisms modifying their own environment, and so the selection pressures on them.

Sources: Jagiello Z, Dylewski L, Tobolka M, Aguirre JI (2019) Life in a polluted world: a global review of anthropogenic materials in bird nests. Environmental Pollution 251: 717 to 722. Suárez-Rodríguez M, López-Rull I, Macías Garcia C (2013) Incorporation of cigarette butts into nests reduces nest ectoparasite load in urban birds. Biology Letters 9: 20120931. Hiemstra A-F, Moeliker CW, Gravendeel B, Schilthuizen M (2023) Bird nests made from anti-bird spikes. Deinsea 21: 17 to 25.

# Scan + Splat Renders

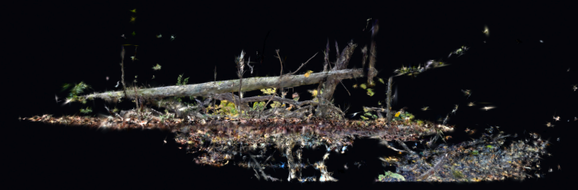
PIN-08

Gaussian splat of the patch (Polycam scan, trained locally, gsplat renders on ink): the fallen utility pole on oaks.

Splat, high orbit B: the patch as an island, the pole diagonal across it →

Splat, atmos, orbit 225° (ink)

Splat, side view (ortho), full frame



Splat, close: the pole's long side · 3.1 m out

Splat, atmos, orbit 180° (ink)

Splat, close: down the length of the pole · 2.5 m out

**Polycam scan; gaussian splat trained locally (COLMAP + Brush); renders: gsplat, on ink.**
Animations (orbits, 360, walk-through): laserpirate.github.io/ett3/a3-groundtruth/animations/

Panels rendered at 600 ppi at their printed size; scan field visit Oct 16 2026.